

Weekly Report

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Project: UAV-Based Traffic Safety & Video Analytics

Week: 7 Mar 2026 - 14 Mar 2026

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Detailed Summary of Work Done During Current Week

A. Zone Alignment Verification

- Verified that predefined polygon zones remain correctly aligned with the road structure throughout the video sequence.
- Conducted additional validation across multiple segments of the UAV traffic video.
- Confirmed that zone boundaries remain spatially consistent and stable across frames.
- Ensured that the predefined zones are suitable for subsequent vehicle trajectory analysis and zone-based behavior evaluation.

B. Semi-Automated Annotation Pipeline Development

Work continued on improving the semi-automated annotation workflow designed to handle the large UAV dataset (~9000 frames).

1. Inference-Based Detection

- Performed object detection inference on video frames using the trained YOLO model.
- Generated bounding boxes along with class labels and confidence scores.
- Verified detection outputs for correctness and consistency across frames.

2. CSV-Based Detection Storage

- Stored inference outputs in a structured CSV format including:
 - Frame number
 - Bounding box coordinates
 - Class label
 - Confidence score
- Validated CSV structure to ensure compatibility with downstream processing and annotation tools.

3. CSV to XML Conversion Refinement

- Continued refining the script used to convert CSV detections into CVAT-compatible XML format.
- Ensured that the XML structure aligns with CVAT's video annotation schema.
- Conducted additional testing to verify that generated XML files can be imported correctly into CVAT.

4. CVAT Annotation Workflow Testing

- Tested the semi-automated annotation workflow using CVAT.
- Verified that pre-generated bounding boxes can be edited, corrected, and refined efficiently.
- Evaluated how the inference-assisted pipeline reduces the amount of manual annotation required for large video datasets.

C. Vehicle Trajectory Extraction Experiments

- Continued development of trajectory extraction logic using centroid computation from bounding box coordinates.
- Generated trajectory points for selected vehicles across longer frame segments.
- Visualized vehicle movement paths on video frames to verify trajectory correctness.
- Observed challenges related to maintaining consistent vehicle identity across frames when vehicles overlap or move close together.

Tasks Planned for Coming Week

A. Annotation Workflow Stabilization

- Finalize the CSV → XML → CVAT annotation pipeline.
- Perform annotation experiments on larger frame segments.
- Evaluate annotation efficiency and scalability for large UAV video datasets.

B. Trajectory Extraction Improvement

- Improve vehicle identity consistency across frames.
- Extend trajectory generation to longer video segments.
- Generate clearer trajectory visualizations for validation.